Muzammal Naseer

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Citations 650, H-index: 8 | ☆ 300

Education and Degrees

2018–2022 **Doctor of Philosophy in Engineering & Computer Science**, *Australian National University (ANU).*

- Ph.D. Advisors: Prof. Fatih Porikli, and Dr. Salman Khan.
- I defended my thesis titled *Novel Concepts and Designs for Adversarial Attacks and Defenses.*
- My research was focused on understanding and improving out-of-distribution generalization of machine learning and deep learning based models. I developed new algorithms for adversarial attack and defense methods for computer vision applications including segmentation, object recognition and detection.
- Multiple chapters of my Ph.D. thesis are also peer-reviewed and published at top machine learning and computer vision venues including NeurIPS, CVPR, ICCV, TNNLS, and TPAMI.

2011–2013 **Master of Science in Electrical Engineering**, *King Fahd University of Petroleum & Minerals (KFUPM)*, average GPA 3.75/4.0.

- I graduated with *first class honors* and defended my thesis titled *The Design of Finite Impulse Response (FIR) Wavefield Extrapolation Filters.*
- Research was conducted with the collaboration of *Saudi Aramco* and the resulting software of my thesis was tested on the active oil well exploration sites. Multiple chapters of my thesis are also peer-reviewed and published at the Society of Exploration Geophysicists.
- I took in-depth courses on mathematics, signal processing, systems identification, and control theory.

2006–2010 **Bachelor of Science in Electrical Engineering**, *University of the Punjab (PU)*, average GPA 3.89/4.0, Gold Medalist.

- I graduated by defending my thesis titled *Spectrum Monitoring & Analysis of Interference in EGSM Networks.* This project was funded by Pakistan Frequency Allocation Board (FAB) and telecommunication network operator company, ZONG.

Academic and Research Experience

Sept. 2020–present **Postdoctoral Researcher**, *Computer Vision Laboratory, MBZUAI UAE*

- Currently, my research is focused on video understanding, adversarial learning, and developing insights by analyzing and visualizing the internal workings of deep neural networks.
- I am supervising Master and Ph.D. students as well as research assistants.
- I am actively participating in teaching including lab supervisions and guest lectures at MBZUAI.

Sept. 2018 – Mar. 2020 **Research Associate**, *Inception Institute of Artificial Intelligence - IIAI, UAE*

- Developed cross-domain adversarial attack and task-generalizable neural purification defense.

2018 **Research Associate**, *Data61, CSIRO, Canberra, Australia*

- Developed task-generalizable transferable adversarial attack for neural networks vision systems.

Sept. 2014 – June. 2017 **Lecturer (Full-Time)**, *University of Hafr Al Batin - subcampus of KFUPM, KSA*

- I joined subcampus of KFUPM, now known as University of Hafr Al-Batin as a full-time lecturer to teach variety of *Electrical Engineering* courses including circuit analysis, analog and digital systems, and programming (C/C^{++}/Assembly). I also mentored student's final year graduate projects.

2013 **Teaching Assistant**, *Department of Electrical Engineering, KFUPM, KSA*

- Assisted in Control Theory and Signal Processing courses.

2009 **Research Assistant**, *Frequency Allocation Board (FAB), Pakistan*

- Worked with FAB team to develop the direction finding system for RF Interferers.

Honors and Awards

2019 Student travel award for Vancouver, Canada at *Neural Information Processing Systems* (NeurIPS).

2017 Postgraduate research scholarship by *ANU, Australia* for the period of three years.

2017 Fee remission merit scholarship by *ANU, Australia* for the period of four years.

2011 Master of science scholarship by the *Ministry of Higher Education*, KSA for the period of two years.

2011 Gold Medal by the *University of the Punjab* (PU) for outstanding performance in B.Sc. degree.

2007-09 University merit scholarship by the *University of the Punjab* (PU) consistently for three years.

Research Funding

2022-2025 **MBZUAI-WIS Program for Collaborative Research in AI**, *UAE*

- **Title**: Biomimetic visual-recognition at low resolution with continuous learning.
- **Amount**: 2.7 Million Dollar.
- **Role**: Work Package Leader

Selected Publications

CVPR 2022 **Oral**, top 5.0%
Self-Supervised Video Transformer.
Kanchana Ranasinghe, **Muzammal Naseer**, Salman Khan, Fahad Shahbaz Khan, Michael Ryoo
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022.

ICLR 2022 **Spotlight**, top 5.0%
On Improving Adversarial Transferability of Vision Transformers.
Muzammal Naseer, Kanchana Ranasinghe, Salman Khan, Fahad Shahbaz Khan, Fatih Porikli
International Conference on Learning Representations (ICLR), 2022.

NeurIPS 2021 **Spotlight**, top 3.0%
Intriguing Properties of Vision Transformers.
Muzammal Naseer, Kanchana Ranasinghe, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, Ming-Hsuan Yang
Advances in Neural Information Processing Systems ((NeurIPS), 2021.

ICCV 2021
On Generating Target Transferable Perturbations.
Muzammal Naseer, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, Fatih Porikli
International Conference on Computer Vision (ICCV), 2021.

ICCV 2021
Orthogonal Projection Loss
Kanchana Ranasinghe, **Muzammal Naseer**, Munawar Hayat, Salman Khan, Fahad Shahbaz Khan
International Conference on Computer Vision (ICCV), 2021.

CVPR 2020 **Oral**, top 5.7%
A Self-supervised Approach for Adversarial Robustness
Muzammal Naseer, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, Fatih Porikli
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2020

NeurIPS 2019
Cross-Domain Transferability of Adversarial Perturbations.
Muzammal Naseer, Salman Khan, Harris Khan, Fahad Shahbaz Khan, Fatih Porikli
Advances in Neural Information Processing Systems ((NeurIPS), 2019.

Selected Journal Publications

TPAMI, 2022
Stylized Adversarial Defense
Muzammal Naseer, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, Fatih Porikli
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2022. [Impact factor: 16.39]

TNNLS 2022
Guidance Through Surrogate: Towards a Generic Diagnostic Attack
Muzammal Naseer, Salman Khan, Fatih Porikli, Fahad Shahbaz Khan
IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2022 [Impact factor: 10.45]

Selected Other Conference Publications

BMVC, 2021
Rich Semantics Improve Few-shot Learning
Mohamed Afham, Salman Khan, Haris Khan, **Muzammal Naseer**, Fahad Khan
The British Machine Vision Conference (BMVC), 2021

WACV 2019
Local Gradients Smoothing: Defense Against Localized Adversarial Attacks
Muzammal Naseer, Salman Khan, Fatih Porikli
Winter Conference on Applications of Computer Vision (WACV), 2019

Computational Skills

Python	I am extensively using python to build machine learning algorithms for the last few years.
Pytorch	It is usually my default choice due to its dynamic nature and object-oriented graph design approach.
Tensorflow	I have used Tensorflow to build different projects including a defense algorithm against unconstrained adversarial attacks.
C/C^{++}	I scored A+ grades in these languages in my B.Sc courses. [Transcript]

Research Supervision

2021–present	Hasmat Shadab Malik, MS student at MBZ University of AI (MBZUAI, UAE)
2021–present	Shahina Kunhimon, Ph.D. student at MBZ University of AI (MBZUAI, UAE)
2020-2021	Kanchana Ranasinghe, Research associate at MBZ University of AI (MBZUAI, UAE)
2020-2021	Mohamed Afham, Research associate at MBZ University of AI (MBZUAI, UAE)

Professional Services

Program Committees

2022	Vision Transformers: Theory and applications, ACCV 2022
2021	Adversarial Robustness In the Real World, ICCV 2021
2020	Adversarial Robustness In the Real World, ECCV 2020

Reviewer

From 2022	Int. Conf. Computer Vision and Pattern Recognition (CVPR)
From 2022	Int. Conf. Advances in Neural Information Processing Systems (NeurIPS)
From 2021	Int. Journals: TPAMI, TIP, and TNNLS

References

Dr. Salman Khan, Associate Professor at MBZ University of AI (MBZUAI, UAE)
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Dr. Fahad Shahbaz Khan, Associate Professor at MBZ University of AI (MBZUAI, UAE)
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Prof. Fatih Porikli, Chief Scientist at Qualcomm, United States
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